

UCL Biodiversity Futures Summer School 2026

Mosquito mapping practical with Dr. Rory Gibb

Exploring data on the abundance of *Aedes albopictus* in Italy



In this short R practical we will use mosquito surveillance data to explore the distribution of *Aedes albopictus*, the Asian tiger mosquito, in Northern and Central Italy. The goal of this practical is to familiarise you with the kinds of tools we can use to rapidly explore and visualise spatial data - tools and skills which are critical for biodiversity and health data science. For this, we will mainly be using spatial data tools, and in particular the *terra* and *sf* packages in R.

The tiger mosquito (*Ae. albopictus*) has gradually colonised Europe over the last half-century, creating a new series of escalating risks for mosquito-borne disease. This species is a competent vector for several viruses that pose major threats to human health, in particular dengue and chikungunya. First found in Italy in the 1990s, it has now established populations across most of the country, and during the last decade several Italian cities have seen significant, sustained outbreaks of dengue and chikungunya (both vectored by this species). In response Italian health authorities have been conducting year-on-year mosquito surveillance (trapping) to understand this species' populations and the factors that create outbreak risks. A large set of surveillance data from several Italian provinces was recently published - you can read the paper here - and this dataset forms the basis for today's practical.

This workbook is structured as a series of code snippets with some reflective questions and exercises interspersed. You can follow the instructions and copy the code into an RStudio script, but also please feel free to explore the data and try out your own ideas.

All the data you'll need for the workshop are on GitHub, in the "pnl_summer_school" repository. Please download the whole folder using download-directory.github.io. Set this folder as your working directory, then all the materials you will need are contained within the "data" subfolder.

```

# check if required packages (deps) are installed and install if not
ip = installed.packages()
deps = c("terra", "dplyr", "magrittr", "sf", "ggplot2", "sf", "rstudioapi", "tidyr", "stringr")
for(d in deps){
  if(!d %in% rownames(ip)){
    print(paste("Installing:", d, sep=" "))
    install.packages(d)
  }
}

# load libraries dependencies
library(terra); library(dplyr); library(magrittr); library(ggplot2); library(sf)
library(rstudioapi); library(tidyr); library(stringr)

# automatically set working directory
# if this doesn't work,
# manually set your working directory to the folder "Week3-Measuring-Env-Effects")
PATH = dirname(rstudioapi::getSourceEditorContext()$path)
setwd(PATH)

```

Understanding the dataset

We will use a subset of a larger mosquito surveillance database that was recently published in *Scientific Data* (Da Re et al, 2024). Our data consist of monthly counts of *Ae. albopictus* eggs, collected using ovitraps, for several months each year between 2010 and 2022. Ovitrap provide a water container in which female mosquitoes lay their eggs, which are subsequently collected and counted. Repeat surveys were conducted monthly at almost 100 unique locations - specified as the column “ID” in the dataset - within five Italian regions (Trento, Emilia-Romagna, Lazio, Tuscany and Veneto). Here, we can use **egg counts as a proxy for mosquito abundance**, and begin to ask what patterns of mosquito abundance look like over space, time and in relation to the environment.

Explore the surveillance dataset. What months of the year do we have data for? What regions do the data come from, and do you notice anything about the different amounts of data between regions? What other information does the data frame contain, in addition to the monthly mosquito counts?

```

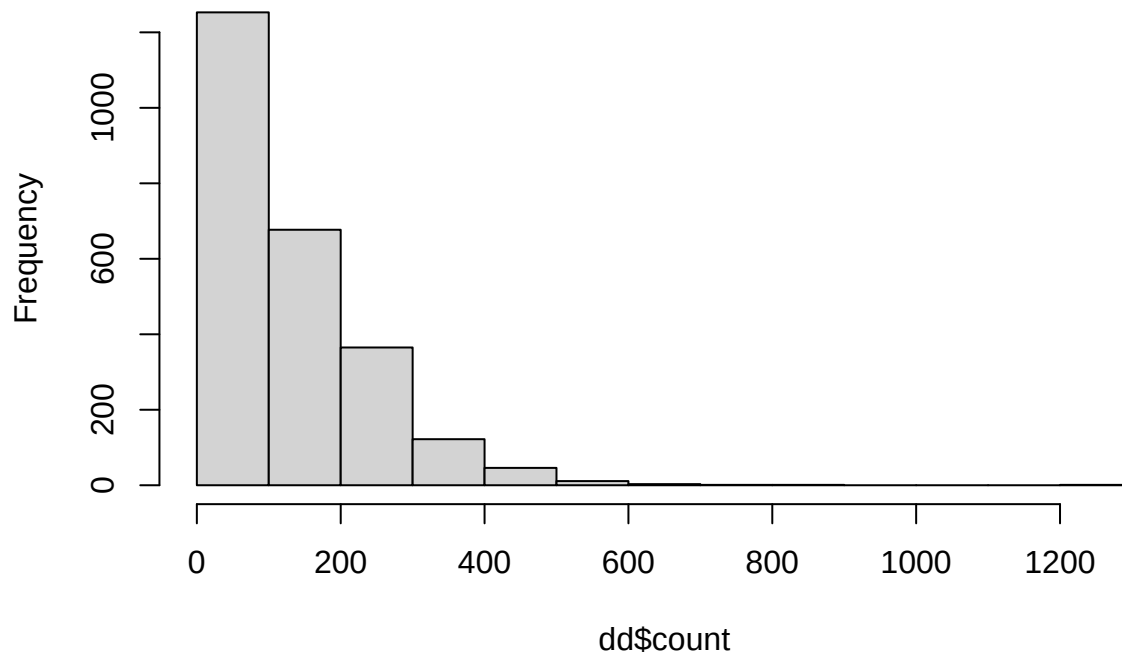
# the mosquito surveillance data are stored as a csv
# mosquito counts are stored in the "count" column
dd = read.csv("./data/vectabundance/italy_vectabundance_summercounts_monthly.csv")

# explore this dataset - you can use the "head" function to look at the top rows of the data
head(dd)

# and you can plot histograms of the numeric variables using hist()
# here is a histogram of counts
# remember, we use the $ operator to select a specific column from a data frame
hist(dd$count)

```

Histogram of dd\$count



```
# you can also generate quick tables of how much data fall in each group
table(dd$month)
table(dd$Region)

# what other variables does the dataset contain?
# plot some histograms and explore them
```

For any data analysis project, we need to understand what exactly the data are, their spatial and temporal distribution, and any additional information that has been collected. These data contain spatial survey locations (in the x and y column) so a useful starting point would be to map them.

Point, line and polygon spatial data can be stored in geographic objects called shapefiles, which contain spatial objects along with rows of metadata describing them. In our dataset, we have a shapefile of Italian regions and another shapefile of the overall Italian border. We can read these into R, and use them as a base for the map we will make.

```
# italian municipalities
shp = sf::st_read("./data/shapefiles/italy_vectabundance_regions.shp")

# italian border (for mapping)
shp_ita = sf::st_read("./data/shapefiles/gadm41_ITA_0.shp")

# take a quick look at the shapefile
# what do you see? R tells you about what this is, and what it contains

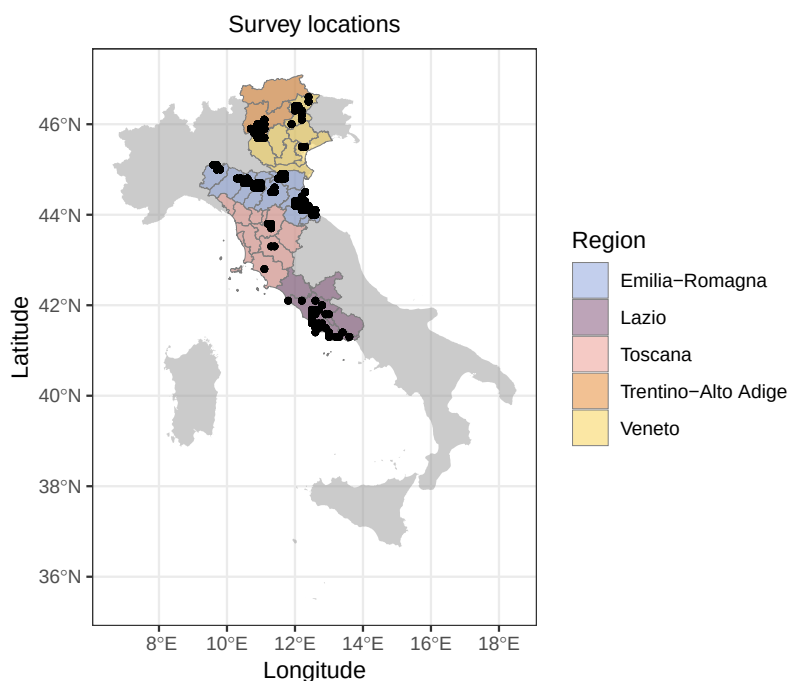
# plot the map
```

```

# we use ggplot, which creates a plot by adding successive levels
# of data as visual features, which are called "geoms"
# so we pass the ggplot function the shapefile data to plot the map borders
# and we pass it the mosquito data to plot the X-Y coordinates of survey locations
map = ggplot() +
  geom_sf(data=shp_ita, color=NA, fill="grey60", alpha=0.5) +
  geom_sf(data=shp, color="grey50", aes(fill=Region), alpha=0.5) +
  geom_point(data = dd, aes(x, y), color='black', size=1)
map

# this map looks a bit basic
# let's make it look nicer by changing the background
# and giving it a title and a nicer colour scale
map = map +
  theme_bw() +
  MetBrewer::scale_fill_met_d(name="Archambault") +
  ggtitle("Survey locations") +
  theme(legend.position="right",
        plot.title=element_text(size=11, hjust=0.5)) +
  xlab("Longitude") + ylab("Latitude")
map

```



Now let's plot a histogram or bar plot of the number of observations across years, and across months. What do you notice - is the sampling relatively even across years and months, or does it vary?

```

# count the number of observations in each year
obs_by_year = dd %>%
  dplyr::group_by(year) %>%
  dplyr::summarise(obs = length(count))

```

```

# plot a bar plot
ggplot(obs_by_year) +
  geom_bar(aes(year, obs), stat="identity", fill="skyblue4") +
  theme_classic() +
  xlab("Year") + ylab("Number of observations")

# 2. same plot but break down by region
# summarise by year and region
obs_by_year = dd %>%
  dplyr::group_by(year, Region) %>%
  dplyr::summarise(obs = length(count))

# use ggplot's "facet_wrap" function to break down into region subplots
ggplot(obs_by_year) +
  geom_bar(aes(year, obs), stat="identity", fill="skyblue4") +
  theme_classic() +
  facet_wrap(~Region) +
  xlab("Year") + ylab("Number of observations") +
  theme(strip.background = element_blank())

```

Before we start analysing data it's also critical to visualise and understand **variation in our variable of interest**. This is stored in the column "count". Looking at the dataset (*hint: you can use R's "head()" function to examine it*) it's clear that we have a form of nested time series data. At each survey location (**denoted by the column "ID"**) we have monthly counts during the summer months (June to September), over multiple years. We call this nested, because we have multiple observations in each location.

We can visualise these time series for some example survey locations, to demonstrate the structure of the data. It's clear that there is a lot of variation in abundance between different locations, and also between years.

```

# subset to a random selection of locations
random_locs = dd %>%
  dplyr::filter(ID %in% c(6777, 8189, 7490, 14199))

# create a "date" column for easier plotting
random_locs$date = as.Date(paste(random_locs$year, random_locs$month, "01", sep="-"))

rl_plot =
  ggplot(random_locs) +
    geom_line(aes(date, count, group=year)) +
    geom_point(aes(date, count, group=year)) +
    theme_classic() +
    facet_wrap(~ID, ncol=1) +
    xlab("Month") + ylab("Ae. albopictus abundance") +
    ggtitle("Aedes albopictus count time-series for example locations") +
    theme(strip.background = element_blank(),
          plot.title=element_text(size=11, hjust=0.5))
rl_plot

```

We can also visualise and compare the distributions of counts across each survey year, for example using boxplots. Are mosquito abundance levels quite similar during most years, or is there interannual variation or long-term trends? (Either `geom_boxplot()` or `geom_violin()` might both be helpful here.)

```

# look at the overall variation across years,
# in each region. what do you notice?
boxplot_by_year = ggplot(dd) +
  geom_boxplot(aes(factor(year), count)) +
  theme_classic() +
  xlab("Year")+ ylab("Ae. albopictus abundance") +
  facet_wrap(~Region) +
  theme(strip.background = element_blank(),
        plot.title=element_text(size=11, hjust=0.5))
boxplot_by_year

```

Mapping the abundance of *Aedes albopictus* over space and time

Next, let's map the distribution of abundances over space and time, to assess whether there are any obvious spatial trends. Since we have multiple counts for each location, we can't effectively show everything on a map at once. One option is to instead visualise the mean count at each location. When you plot this map, what do you notice about how *Ae. albopictus* is distributed geographically? What do you think could be causing this?

```

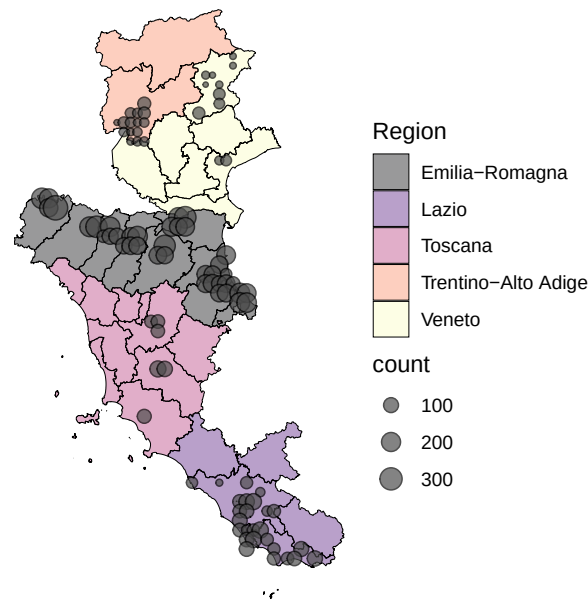
# calculate mean count per location across all surveys
mean_counts = dd %>%
  dplyr::group_by(ID) %>%
  dplyr::summarise(count = mean(count, na.rm=TRUE),
                  x = head(x, 1),
                  y = head(y, 1))

# plot the map of average differences between sampling locations
mc_map = ggplot(mean_counts) +
  geom_sf(data=shp, color="black", aes(fill=Region), alpha=0.4) +
  theme_void() +
  geom_point(aes(x, y, size=count), fill="grey30", alpha=0.7, pch=21) +
  scale_color_viridis_c(name="Mean\nsummer\ncount") +
  scale_fill_viridis_d(option="magma") +
  scale_size(range=c(0.8, 5)) +
  ggtitle("Mean Ae. albopictus count per sampling locale") +
  theme(plot.title=element_text(size=11, hjust=0.5))

mc_map

```

Mean Ae. albopictus count per sampling locale



Finally, we can also plot a series of maps to look at how the distribution might have changed over time. Here, let's combine the observations from a few years at a time, and plot multiple maps, one for each time period.

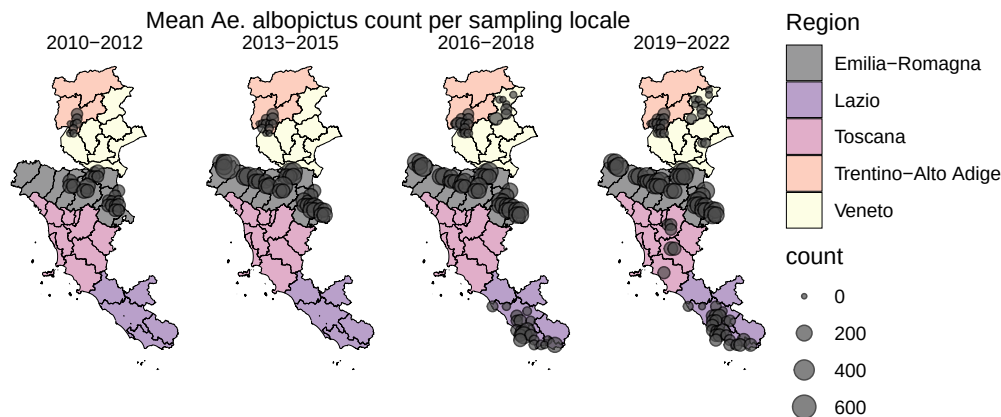
What patterns do you notice, if any?

```
# create "year group" field
year_grp = data.frame(
  year = 2010:2022,
  year_grp = c(rep("2010-2012", 3), rep("2013-2015", 3), rep("2016-2018", 3), rep("2019-2022", 4))
)

# calculate mean count per location per year group
mean_counts = dd %>%
  dplyr::left_join(year_grp) %>%
  dplyr::group_by(ID, year_grp) %>%
  dplyr::summarise(count = mean(count, na.rm=TRUE),
    x = head(x, 1),
    y = head(y, 1))

# plot the map of average differences between sampling locations
mc_map = ggplot(mean_counts) +
  geom_sf(data=shp, color="black", aes(fill=Region), alpha=0.4) +
  theme_void() +
  geom_point(aes(x, y, size=count), fill="grey30", alpha=0.7, pch=21) +
  scale_color_viridis_c(name="Mean\nsummer\ncount") +
  scale_fill_viridis_d(option="magma") +
  scale_size(range=c(0.8, 5)) +
  ggtitle("Mean Ae. albopictus count per sampling locale") +
  theme(plot.title=element_text(size=11, hjust=0.5)) +
  facet_wrap(~year_grp, nrow=1)
```

```
mc_map
```



Investigating the relationship between temperature and monthly *Aedes albopictus* abundance

Our maps indicate that *Ae. albopictus* appears to be particularly abundant in the Emilia-Romagna region, and less so further to the north and south. Italy is a very mountainous country so it's possible that local climatic factors are playing a role. The dataset contains a column - "temperature_c" - which contains the monthly average temperature during the month preceding surveys. Let's start exploring whether temperature could play a role.

Let's examine the correlation between monthly temperature and *Ae. albopictus* egg count.

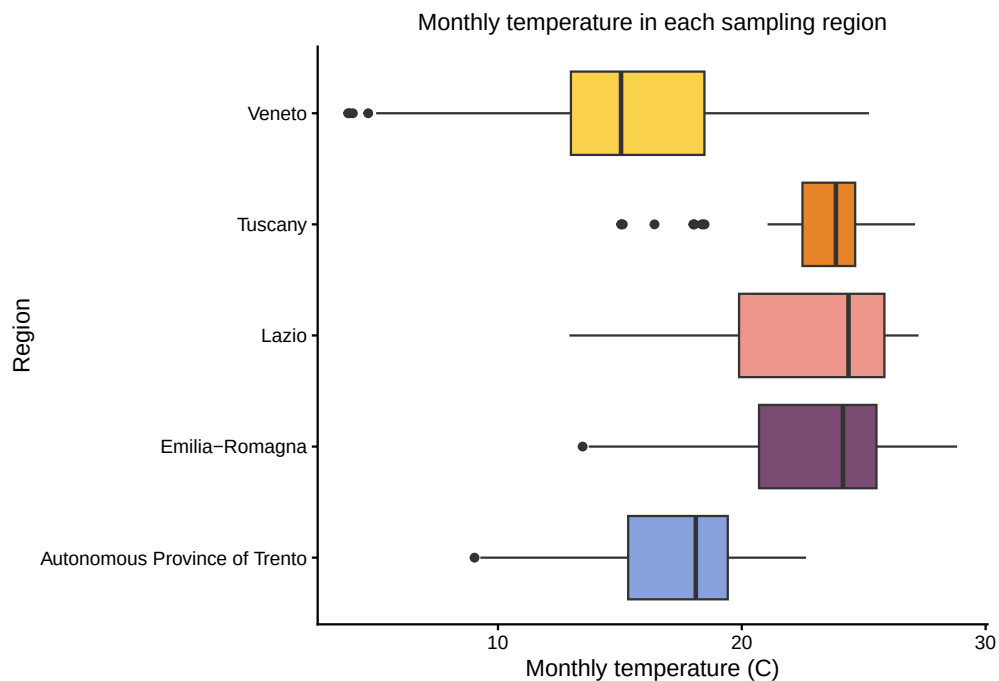
The code below plots a scatter plot of the relationship between temperature and egg counts. Discuss this as a group and think about what could be going on here. Is there visual evidence of a relationship?

```
# scatter plot of temp vs counts
ggplot(dd) +
  geom_point(aes(temperature_c, count), color="skyblue4", alpha=0.7, size=0.8) +
  xlab("Monthly temperature (C)") + ylab("Ae albopictus egg count") +
  theme_bw() +
  theme_classic()
```

We can also use boxplots or scatter plots to examine differences in temperature between the different regions where our data were collected. What do you notice about the regional differences in temperature? Do you think these could be playing a role?

```
# scatter plot of temp vs counts
ggplot(dd) +
```

```
geom_boxplot(aes(Region, temperature_c, fill=Region)) +
theme_classic() +
xlab("Region") + ylab("Monthly temperature (C)") +
MetBrewer::scale_fill_met_d(name = "Archambault") +
ggtitle("Monthly temperature in each sampling region") +
theme(plot.title=element_text(size=11, hjust=0.5), legend.position="none") +
coord_flip()
```



Suggested extra extension exercises

In the full modules we run on these topics, we go into much more depth into how we can use statistics to infer the strength and shape of the relationships between climate, environment and mosquito densities. If you're interested to explore further, the dataset includes columns for another important variable that could influence mosquito abundance: precipitation, which creates the bodies of standing water that most species need to lay their eggs ("breeding sites"). If you are interested, look at the pattern of precipitation across regions and if there is a relationship between rainfall levels and mosquito abundance. Is there evidence of a relationship?